



- 2 (a) With reference to thermal energy, state what is meant by two objects being in thermal equilibrium.

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 [1]

- (b) Two cylinders X and Y each contain a sample of an ideal gas. The samples are in thermal equilibrium with each other.

X has a volume of 0.0260 m^3 and contains 0.740 mol of gas at a pressure of $1.20 \times 10^5 \text{ Pa}$. Y has a volume of 0.0430 m^3 and contains gas at a pressure of $2.90 \times 10^5 \text{ Pa}$. Data for the two cylinders are shown in Fig. 2.1.



Fig. 2.1

- (i) Show that the temperature of the gas in X is 234°C .

[3]

- (ii) Determine the number N of molecules of the gas in Y. Explain your reasoning.

$N =$ [3]





- (iii) The gas in X consists of molecules that each have a mass that is four times the mass of a molecule of the gas in Y.

Explain how the root-mean-square (r.m.s.) speed of the molecules in X compares with the r.m.s. speed of the molecules in Y.

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..... [3]

[Total: 10]

